DIEGO MAGANA

Los Angeles, CA • diegomagana@gmail.com • (480) 544-2890 • github.com/diego-magana • diego-magana.github.io

Principal Engineer with ~10 years across data science, ML, and engineering, now pursuing an MSCS (AI · GPA 3.88). Builds deep-learning systems from first principles — autograd, language models, transformers — and investigates their internal mechanisms through causal interpretability. Focused on mechanistic interpretability, alignment, and building systems whose behavior can be explained, not just measured.

EDUCATION

Johns Hopkins University — Baltimore, MD

Master of Science, Computer Science (AI) — in progress · GPA 3.88

Coursework: Data Structures (A+) · Programming in Java (A) · Calculus I (A) · Calculus II (A-)

Amherst College — Amherst, MA

B.A., Environmental Studies (2017) · NCAA Division III Men's Basketball — NESCAC Champion, NCAA Semifinalist (2014)

ML PROJECTS

gpt — Mechanistic Interpretability of a Transformer · github.com/diego-magana/gpt

- Found that a single attention head carries a character-level transformer's next token prediction and isolated it with convergent causal evidence — head ablation (+0.89 nats, 7× the next head) and single-head activation patching (~0.51 recovery, 5× the next) — on cache, ablation, and patching instrumentation built from scratch.
- Confirmed the mechanism with a pairwise-ablation redundancy test and replicated it across five training seeds, finding it robust but its head index an initialization artifact; tested the boundary with a corruption-distance sweep and reported the architecture's limits explicitly rather than over-claiming generality.

micrograd — Autograd Engine from First Principles · github.com/diego-magana/micrograd

- Built a scalar-valued reverse-mode autodiff engine from scratch — dynamic graph via operator overloading — with gradients verified against PyTorch to 1e-5 across all operators; trained an MLP (337 params) to 100% accuracy on make_moons.
- Packaged as a tested library with CI, extended beyond the reference implementation to sigmoid and ReLU activations and multi-class classification on 3-class spirals.

makemore — Character-Level Language Model Series · github.com/diego-magana/makemore

- Implemented the full character-level progression — bigram → MLP (Bengio 2003) → WaveNet (77K params, 2.01 validation NLL) — including a from-scratch backprop variant matching PyTorch autograd to within ~0.001 nats.
- Authored an original probing analysis of the trained model's representations — embedding geometry, gradient-based context sensitivity, and single-unit ablation — with explicit reporting of correlational vs. causal limits.

EXPERIENCE

Raxwell Industries, Irvine, CA — Jan '23 – Present

Principal Engineer (Contract) — *Oct '25 – Present*

- Built a component-separated RAG evaluation harness benchmarking open-weights models (Llama 3.1, Mistral) against managed copilots — scoring retrieval, faithfulness, and numerical accuracy as separate gates — producing the build-vs-buy cost model.
- Instrumented the pipeline end-to-end (per-component tracing, CI regression gate), diagnosing failures by isolating retrieval miss from generation drift.

Chief Information Officer — *Mar '24 – Oct '25*

- Owned end-to-end technology strategy and execution through a growth phase as the senior-most technical authority — Azure cloud architecture and enterprise data infrastructure.
- Delivered enterprise system integrations across finance, supply chain, and EDI — connecting Business Central with external trading-partner and operational systems as the backbone of daily operations.

Principal Engineer (Contract) — *Jan '23 – Mar '24*

- Architected the company's Azure migration and built the data platform and ML/analytics infrastructure still in production today.
- Built ETL pipelines and REST API integrations feeding the production data platform.

Mavco LLC, Scottsdale, AZ — Aug '17 – Jan '23

Senior Data Scientist — *Aug '20 – Jan '23*

- Established the company's first analytics and ML function, scaling the modeled catalog from hundreds of SKUs to tens of thousands via automated pipelines.
- Owned the full ML lifecycle for forecasting, pricing, and classification systems — data engineering through production deployment — driving a ~10-point lift in adjusted gross margin and informing seven-figure capital allocation.

Data Scientist — *Aug '17 – Aug '20*

- Built ML pipelines for sales and pricing prediction and developed financial models informing executive strategy.

TECHNICAL SKILLS

- **Deep Learning:** PyTorch, transformer architectures, LLM pretraining & fine-tuning, language modeling
- **Interpretability & Analysis:** activation/residual/head patching, logit lens, head ablation; causal vs. correlational evaluation
- **ML Systems & Evaluation:** RAG evaluation harnesses, component-separated metrics, CI regression gating
- **Machine Learning:** scikit-learn, time-series forecasting, classification, statistical modeling
- **Infrastructure & MLOps:** Docker, GitHub Actions (CI/CD), Modal, W&B, Azure (AI Foundry model deployment, integration)
- **Languages:** Python, Java, SQL, C++